

RUBAISHA ARIF
CT-25067

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OOP LAB 3

Example 1: Default Constructor

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Construct { 4 public: 5 int a, b; 6 Construct() { 7 a = 10; 8 b = 20; 9 } 10 }; 11 int main() { 12 Construct c; 13 cout << "a: " << c.a << endl; 14 cout << "b: " << c.b << endl; 15 return 0; 16 } 17</pre>	<pre>a: 10 b: 20 === Code Execution Successful ===</pre>

Example 2: Default Constructor

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Wall { 4 private: 5 double length; 6 public: 7 Wall() { 8 length = 5.5; 9 cout << "Creating a wall." << endl; 10 cout << "Length = " << length << endl; 11 } 12 }; 13 int main() { 14 Wall wall1; 15 return 0; 16 } 17 18</pre>	<pre>Creating a wall. Length = 5.5 === Code Execution Successful ===</pre>

Example 1: Parameterized Constructor

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class Point { 5 int x, y; 6 public: 7 Point(int x1, int y1) { 8 x = x1; 9 y = y1; 10 } 11 int getX() { 12 return x; 13 } 14 int getY() { 15 return y; 16 } 17 }; 18 int main() { 19 Point p1(10, 15); 20 21 cout << "p1.x = " << p1.getX() 22 << ", p1.y = " << p1.getY() << endl; 23 return 0; 24 }</pre>	<pre>p1.x = 10, p1.y = 15 === Code Execution Successful ===</pre>

Example 2: Parameterized Constructor

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class Wall { 5 6 double length; 7 double height; 8 public: 9 10 Wall(double len, double hgt) { 11 length = len; 12 height = hgt; 13 } 14 double calculateArea() { 15 return length * height; 16 } 17 }; 18 19 int main() { 20 21 Wall wall1(10.5, 8.6); 22 Wall wall2(8.5, 6.3); 23 cout << "Area of Wall 1: " << wall1 24 .calculateArea() << endl; 25 cout << "Area of Wall 2: " << wall2 26 .calculateArea() << endl; 27 28 return 0; 29 }</pre>	<pre>Area of Wall 1: 90.3 Area of Wall 2: 53.55 === Code Execution Successful ===</pre>

Example 1: Copy Constructor

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class Example { 5 int a, b; 6 public: 7 8 Example(int x, int y) { 9 a = x; 10 b = y; 11 cout << "\nI am Constructor"; 12 } 13 Example(const Example& obj) { 14 a = obj.a; 15 b = obj.b; 16 cout << "\nI am Copy Constructor"; 17 } 18 19 void Display() { 20 cout << "\nValues: " << a << "\t" << b; 21 } 22 }; 23 int main() { 24 Example Object(10, 20); 25 Example Object2(Object); 26 Example Object3 = Object; 27 Object.Display(); 28 Object2.Display(); 29 Object3.Display(); 30 return 0; 31 }</pre>	<pre>I am Constructor I am Copy Constructor I am Copy Constructor Values: 10 20 Values: 10 20 Values: 10 20 === Code Execution Successful ===</pre>

Example 2: Copy Constructor

main.cpp	Output
<pre>1 #include <iostream> 2 #include <cstdlib> 3 using namespace std; 4 class Wall { 5 double* length; 6 double* height; 7 public: 8 Wall() { 9 length = nullptr; 10 height = nullptr; 11 } 12 Wall(double len, double hgt) { 13 length = new double; 14 height = new double; 15 if (length != nullptr && height != nullptr) 16 { 17 *length = len; 18 *height = hgt; 19 } else { 20 exit(1); 21 } 22 } 23 Wall(Wall& obj) { 24 length = new double; 25 height = new double; 26 27 *length = *(obj.length); 28 *height = *(obj.height); 29 } 30 void set_values(double len, double hei) { 31 *length = len;</pre>	<pre>Area of Wall 1: 90.3 Area of Wall 2: 53.55 === Code Execution Successful ===</pre>

main.cpp	Output
<pre> 30 *length = len; 31 *height = hei; 32 } 33 double calculateArea() { 34 return (*length) * (*height); 35 } 36 ~Wall() { 37 delete height; 38 delete length; 39 } 40 }; 41 int main() { 42 Wall wall1(10.5, 8.6); 43 Wall wall2 = wall1; 44 45 cout << "Area of Wall 1: " << wall1 46 .calculateArea() << endl; 47 cout << "Area of Wall 2: " << wall2 48 .calculateArea() << endl; 49 wall2.set_values(2.3, 1.5); 50 51 cout << "After changing values for Wall 2" << 52 endl; 53 cout << "Area of Wall 1: " << wall1 54 .calculateArea() << endl; 55 cout << "Area of Wall 2: " << wall2 56 .calculateArea() << endl; 57 return 0; 58 } </pre>	<pre> Area of Wall 1: 90.3 Area of Wall 2: 53.55 === Code Execution Successful === </pre>

Destructor

main.cpp	Output
<pre> 1 #include <iostream> 2 using namespace std; 3 class HelloWorld { 4 public: 5 HelloWorld() { 6 cout << "Constructor is called" << endl; 7 } 8 ~HelloWorld() { 9 cout << "Destructor is called" << endl; 10 } 11 void display() { 12 cout << "Hello World!" << endl; 13 } 14 }; 15 int main() { 16 HelloWorld obj; 17 obj.display(); 18 return 0; 19 } </pre>	<pre> Constructor is called Hello World! Destructor is called === Code Execution Successful === </pre>

EXERCISE Q1

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Copy { 4 int x,y; 5 public: 6 Copy() { //constructor 7 x = 10; 8 y=20; 9 cout<<"constructor invoked!\n"; 10 } 11 Copy(const Copy &c) { // Copy constructor 12 x = c.x; 13 y = c.y; 14 cout<<"Copy constructor invoked!\n"; 15 } 16 void display() { 17 cout << "x: " << x << endl; 18 cout << "y: " << y << endl; 19 } 20 ~Copy() 21 { 22 cout<<"destructor invoked!\n"; 23 }; 24 int main() { 25 Copy c1; 26 Copy c2 = c1; 27 Copy c3 = c1; 28 c1.display(); 29 c2.display(); 30 c3.display(); 31 return 0;}</pre>	<pre>constructor invoked! Copy constructor invoked! Copy constructor invoked! x: 10 y: 20 x: 10 y: 20 x: 10 y: 20 destructor invoked! destructor invoked! destructor invoked! === Code Execution Successful ===</pre>

Q2

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class tollbooth { 5 6 int Tcar; // total number of cars 7 double Tmoney; // total amount of money collected 8 public: 9 tollbooth() { 10 Tcar = 0; 11 Tmoney = 0; 12 } 13 void payingCar() { 14 Tcar++; 15 Tmoney += 0.50; 16 } 17 void display() { 18 cout << "Total cars: " << Tcar << endl; 19 cout << "Total money: " << Tmoney << endl; 20 } 21 }; 22 int main() { 23 tollbooth t1; 24 t1.display(); 25 t1.payingCar(); 26 t1.payingCar(); 27 t1.payingCar(); 28 t1.display(); 29 return 0;}</pre>	<pre>Total cars: 0 Total money: 0 Total cars: 3 Total money: 1.5 === Code Execution Successful ===</pre>

Q3

```
main.cpp
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class bookType {
6     string title;
7     string author[4];
8     int Numauthor;
9     string publisher;
10    string ISBN;
11    float price;
12    int copies, year;
13 public:
14    bookType() { //constructor
15        title = " ";
16        Numauthor = 0;
17        publisher = " ";
18        ISBN = " ";
19        price = 0;
20        copies = 0;
21        year = 0;
22        for(int i=0; i<4; i++)
23            { author[i] = " "; }
24    }
25
26    void setTitle(string t) { title = t; }
27    string getTitle() { return title; }
28    bool checkTitle(string t) {
29        if(title == t)
30            return true;
31        else
32            return false;
33    }
34    void setcopies(int c) {
35        if(c > 0)
36            copies = c; }
37    void updatecopies(int c) {
38        if(c > 0)
39            copies += c;
40    }
41    int getcopies() { return copies; }
42
43    void setAuthors(string Author[], int num) {
44        if (num > 4)
45            num = 4;
46        else
```

```
Enter title(without spaces)
HarryPotter
Title : HarryPotter
Authors: J.K, Rowling, xyz,
Publisher : Bloomsbury
ISBN : 978-07475
Price : 25
copies in stock : 50
year of publication : 2000
Enter ISBN
4567
book not found

=== Code Execution Successful ===
```

```
main.cpp
46     else
47         Numauthor = num;
48     for (int i = 0; i < num; i++) {
49         author[i] = Author[i];
50     }
51 }
52 void printAuthors() {
53     for (int i = 0; i < Numauthor; i++) {
54         cout << author[i] << " ";
55     }
56     cout<<endl; }
57 int getAuthorCount() { return Numauthor; }
58 void setPublisher(string pub) { publisher = pub; }
59 string getPublisher() { return publisher; }
60 void setISBN(string isbn) { ISBN = isbn; }
61 string getISBN() { return ISBN; }
62 bool checkISBN(string isbn) { return ISBN == isbn; }
63 void setPrice(float p) {
64     if (p>0)
65         price = p; }
66 float getPrice() { return price; }
67 void setyear(int y)
68 {
69     if (y>0)
70         year = y; }
71 int getyear()
72 { return year; }
73 void printBookDetails() {
74     cout << "Title : " << title << endl;
75     cout << "Authors: ";
76     printAuthors();
77     cout << "Publisher : " << publisher << endl;
78     cout << "ISBN : " << ISBN << endl;
79     cout << "Price : " << price << endl;
80     cout << "copies in stock : " << copies << endl;
81     cout<<"year of publication : "<<year<<endl;
82 }
83 };
84 int main() {
85     bookType book[100];
86     int count = 0;
87     string tsearch; //search by title
88     string ISBNsearch;
89
90     string a1[2] = {"Dennis Ritchie", "Brian Kernighan"}; //book1
91     book[count].setTitle("C Programming");
```

```
Enter title(without spaces)
HarryPotter
Title : HarryPotter
Authors: J.K, Rowling, xyz,
Publisher : Bloomsbury
ISBN : 978-07475
Price : 25
copies in stock : 50
year of publication : 2000
Enter ISBN
4567
book not found

=== Code Execution Successful ===
```

main.cpp

Share

Run

```
83 }
84 int main() {
85     bookType book[100];
86     int count = 0;
87     string tsearch; //search by title
88     string ISBNsearch;
89
90     string a1[2] = {"Dennis Ritchie", "Brian Kernighan"}; //book1
91     book[count].setTitle("C Programming");
92     book[count].setAuthors(a1, 2);
93     book[count].setPublisher("Prentice Hall");
94     book[count].setISBN("9780131103627");
95     book[count].setPrice(2500);
96     book[count].setcopies(10);
97     book[count].setyear(1997);
98     count++;
99     string a2[3] = {"J.K.", "Rowling", "xyz"}; //book2
100    book[count].setTitle("HarryPotter");
101    book[count].setAuthors(a2, 3);
102    book[count].setPublisher("Bloomsbury");
103    book[count].setISBN("978-07475");
104    book[count].setPrice(25.00);
105    book[count].setcopies(50);
106    book[count].setyear(2000);
107    count++;
108    cout<<"Enter title(without spaces)\n";
109    cin>>tsearch;
110    bool flag=false;
111    for (int i = 0; i < count; i++) {
112        if (book[i].checkTitle(tsearch))
113        {
114            book[i].printBookDetails();
115            flag = true;
116        }
117        if(flag==false)
118            cout<<"book not found\n";
119    }
120    cout<<"Enter ISBN\n";
121    cin>>ISBNsearch;
122    bool flag=false;
123    for (int i = 0; i < count; i++) {
124        if (book[i].checkISBN(ISBNsearch)) {
125            book[i].printBookDetails();
126            flag = true; }
127    }
128    if(flag==false)
129        cout<<"book not found\n";
130    return 0; }
```

Output

Enter title(without spaces)
HarryPotter
Title : HarryPotter
Authors : J.K. Rowling, xyz.
Publisher : Bloomsbury
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